

SATQUERY AI — REMOTE SENSING ANALYSIS REPORT

ISRO Problem Statement 26167 • Mission ID: 8b7f654f • Generated: 15:52:39.467Z

Natural Query:	What changed between these two images?
Classified Task:	CHANGE_VQA
Calibrated Confidence:	90% (Change Vector & Semantic Alignment)
Specialist Models:	SatQuery-Change-CVA-v1, SatQuery-Change-VQA-v1
Total Latency:	539.0 ms

GROUNDING AI INTERPRETATION

Analysis of the bi-temporal imagery reveals a total surface change of 18.41%. The predominant dynamic is Vegetation recovery / Agricultural growth.

INPUT SENSOR METADATA

Filename	Modality	Dimensions	Bands	GeoTIFF CRS
sample_change_2024_t	OPTICAL	512x512	3	None (Benchmark)
sample_change_2026_t	OPTICAL	512x512	3	None (Benchmark)

OBSERVABLE AGENTIC EXECUTION TRACE

#	Phase / Event	Observable Details	Time (m)
1	Query Ingestion	Received natural language query: 'What changed between these two images?' with 2 uploaded image(s).	0.0
2	Input Validation	Verified 2 image file(s). Format and size limits satisfied.	0.0
3	Raster Metadata Extraction: sample_change_2024_t1.png	Extracted dimensions (512x512), 3 band(s). Format: Standard Benchmark Image (PNG/JPEG)	44.1
4	Raster Metadata Extraction: sample_change_2026_t2.png	Extracted dimensions (512x512), 3 band(s). Format: Standard Benchmark Image (PNG/JPEG)	58.0
5	Modality Identification	Detected sensor modalities: ['OPTICAL', 'OPTICAL'].	0.0
6	Query Classification → CHANGE_VQA	Bi-temporal pair with specific analytical change query -> Routed to Change-grounded VQA Specialist.	0.1
7	Model Dispatch	Dispatched task to specialist: 'satquery-change-vqa-v1' on device 'cpu'.	0.0
8	Specialist Execution Completed	Inference finalized in 436.1ms. Evidence artifacts compiled.	436.2
9	Result Fusion & Calibration	Derived grounded response with calibrated confidence 90% (Change Vector & Semantic Alignment).	0.0

